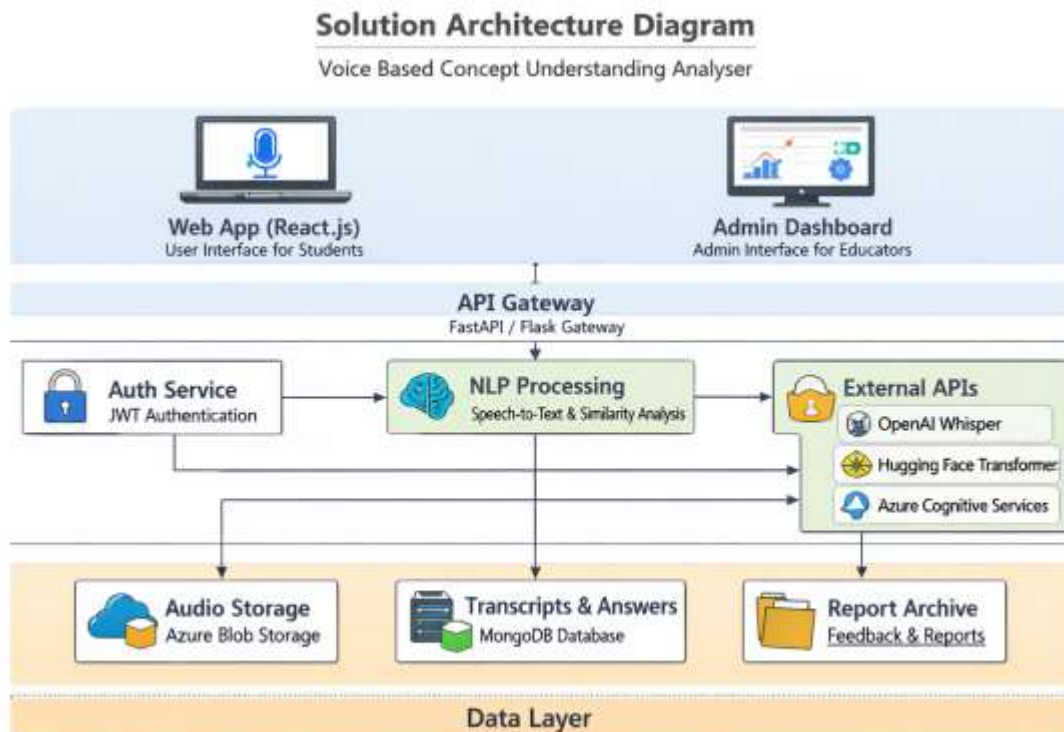


Solution Architecture

Date	15 July 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

A high-level representation of the system architecture is shown below, divided into **Presentation Layer**, **Application Layer**, and **Data Layer**.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for recording/uploading audio and viewing feedback reports.	React.js, HTML, CSS, Bootstrap
API Gateway	Manages request routing between frontend and backend services.	FastAPI / Flask
Microservices	Handles speech-to-text, NLP preprocessing, similarity scoring, and feedback generation.	Python, Whisper API, HuggingFace Transformers
Database	Stores audio, transcripts, reference answers, and feedback reports.	MongoDB, Azure Blob Storage